

Nicholas Mirchandani

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Education

Chapman University

Bachelor of Science:

- Majors: Computer Science & Computer Engineering
- Minor: Game Development

GPA: 3.775

Orange, CA
Class of 2022

Work Experience

Software Engineer II

SpaceX (Starlink) - Redmond, WA

June 2022 - Present

- Primary developer on gateway software for world's fastest satellite internet (Community Gateway)
- Restructured end user metrics calculations to better quantify user experience, filtering out expected issues.
- Generated SQL queries for monitoring trends and triaging issues.
- Generated hardware abstraction layers to use for multiple vehicle generations.
- Wrote high performance, low latency C++ for handling many requests reliably.
- Designed and implemented large scale reworks to existing systems.
- Utilized test frameworks to build lasting regression protection and conduct pre-merge verification.
- Wrote documentation, highlighting various software-hardware interactions that make Starlink possible
- Mentored teammates, allowing management to distribute tasks amongst the entire team.

Undergraduate Research Assistant

Chapman University - Orange, CA

December 2020 - May 2022

- Investigated how to create flexible, large-scale photonic circuits for functional applications.
- Self-taught optoelectronics, watching Cornell's ECE 4300 lectures and videos from Jordan Edmunds.
- Researched using harmonics from Epsilon Near Zero materials for Phase-Shift Keying signal regeneration
- Presented at SPIE Photonics West 2022

Projects

Squishable Sponge

May 2025

- OpenGL Simulation of a Squishable Sponge, capable of absorbing water and squeezing it back out.
- Built on GLFW, GLAD, GLM, and stb_image to get the core rendering and 3D transformations.
- Built particle system from scratch to generate and scatter water particles, complete with object pooling.
- Full source code available on GitHub at <https://github.com/nicholasmirchandani/SquishableSponge>

Fugue

April 2019 - June 2020

- A game made with Unity, designed to showcase the synergy between games and music.
- Lead programmer, bringing team's ideas and designs to life.
- Experimented with shader technologies to create stunning visuals
- Fugue is available at <https://brandoncluff.itch.io/fugue>

AI For Robotics: Transfer Team

November 2021 - December 2021

- Team Lead on class project where we programmed a robot dog to find, collect, and deposit foam bricks.
- Planned out the general algorithm for our solution, broken into pieces to be parallelizable.
- Assigned tasks to different members, adapting based on individual development velocity.
- Hopped in with technical assistance as needed, ensuring the solution properly came together.

Skills

Programming: C, C++, OpenGL, GLSL, SQL, Git, C#, Python, Java, Bazel, Bash

Electronics: Soldering, Digital Logic, RTL Design

Concepts: Data Structures/Algorithms, Object Oriented Programming, Operating Systems, High Performance Computing, Network Security, Computer Organization/Architecture, Algorithm Analysis.

Mathematics: Digital Signal/Image Processing, Discrete Mathematics, Linear Algebra, Multivariable Calculus